



Report
Of
ARTIFICIAL INTELLIGENCE INTERNSHIP

Submitted
To
**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE

By
Roll Number of Student: 1557
Name of Student: **Hardik Rathour**
Batch: 2025-29
Semester: 3rd Semester Section: 2CSE16
2026-27

CANDIDATE’S DECLARATION

I, **Hardik rathour** , do hereby declare that the internship report titled “ARTIFICIAL INTELLIGENCE INTERNSHIP” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: **Hardik rathour**

Roll No.: **1557**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: Hardik rathour . .

Roll No.:1557

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INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP

This certificate is awarded to
Hardik Rathour

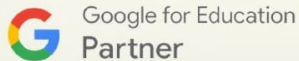
From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Artificial Intelligence Intern

Conducted from 30/06/2026 to 30/07/2026



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, belonging to Dr. Anurag Shrivastava.

Dr. Anurag Shrivastava
Program Manager

4. PROJECT DESCRIPTION

4.1 Introduction

This internship report documents the Artificial Intelligence internship undertaken at Codec Technologies Pvt. Ltd. from 01/06/2026 to 02/07/2026. The internship designation was Artificial Intelligence Intern. The documented work includes two projects: Handwritten Digit Recognizer and Twitter Sentiment Analysis.

4.2 Organization Profile

According to the internship offer letter, Codec Technologies is described as a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. The offer letter states that Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries.

Internship details: Designation – Artificial Intelligence Intern; Location & Mode – Pan India, Hybrid; Duration – 01/06/2026 to 02/07/2026; Reporting – Assigned Project Head(s).

4.3 Problem Statement

Project 1 – Handwritten Digit Recognizer: The problem is to automatically recognize handwritten digits from image data. The project uses the MNIST dataset and a Convolutional Neural Network (CNN) to classify digits from 0 to 9.

Project 2 – Twitter Sentiment Analysis: The problem is to determine the sentiment of Twitter posts and classify them into Negative, Neutral, or Positive categories using Natural Language Processing (NLP) and machine learning.

4.4 Project Objectives

- To understand and implement a CNN-based image classification workflow.
- To preprocess and normalize image data and train a model on MNIST.
- To understand text preprocessing and TF-IDF feature extraction.
- To build a Logistic Regression model for three-class sentiment classification.
- To evaluate model performance using accuracy, classification reports, and confusion matrices.
- To gain practical exposure to Python-based AI/ML development.

4.5 Scope of the Project

The scope covers two practical AI/ML applications. The handwritten digit project focuses on image classification using deep learning, while the sentiment

analysis project focuses on text classification using NLP and classical machine learning. The repositories provide model results, prediction examples, and directions for future enhancement.

4.6 Technologies and Tools Used

Project	Technologies / Tools	Purpose
Handwritten Digit Recognizer	Python; TensorFlow; Keras; NumPy; Matplotlib; CNN; MNIST	Image preprocessing, CNN training, digit classification and evaluation.
Twitter Sentiment Analysis	Python; Pandas; Scikit-learn; Matplotlib; NLP; TF-IDF; Logistic Regression; Google Colab	Text cleaning, feature extraction, sentiment classification and evaluation.

4.7 System Architecture

4.7.1 Handwritten Digit Recognizer

The CNN architecture described in the project repository consists of: (1) Convolutional Layer – 32 filters, (2) Max Pooling Layer, (3) Convolutional Layer – 64 filters, (4) Max Pooling Layer, (5) Flatten Layer, (6) Dense Layer – 128 neurons, (7) Dropout Layer, and (8) Output Layer – 10 classes.

Figure 4.1: Handwritten Digit Recognizer flow: Input image → Convolution → Max Pooling → Convolution → Max Pooling → Flatten → Dense → Dropout → 10-class Output.

4.7.2 Twitter Sentiment Analysis

Figure 4.2: Twitter Sentiment Analysis flow: Twitter Dataset → Data Cleaning → Remove Missing Values → Text Preprocessing → TF-IDF Vectorization → Train/Test Split → Logistic Regression → Predictions → Evaluation → Sentiment Classification.

The sentiment classes are Negative (-1), Neutral (0), and Positive (1).

4.8 Methodology

4.8.1 Handwritten Digit Recognizer

- Load the MNIST dataset.
- Preprocess and normalize image data.
- Train the CNN using TensorFlow/Keras.
- Evaluate the trained model on unseen test data.
- Generate digit predictions and a confusion matrix.
- The repository reports 99.19% test accuracy after 5 epochs.

4.8.2 Twitter Sentiment Analysis

- Clean the Twitter sentiment dataset and remove missing values.
- Perform text preprocessing.
- Convert text into numerical features using TF-IDF.
- Use a train/test split for model evaluation.
- Train a Logistic Regression classifier.
- Generate predictions and evaluate using accuracy, classification report and confusion matrix.
- The repository reports 85.46% test accuracy on 32,593 test samples.

4.9 Expected Outcomes

The expected outcomes are practical understanding of AI/ML workflows, experience with model training and evaluation, and the ability to apply appropriate techniques to image and text classification problems.

The documented results include 99.19% test accuracy for the Handwritten Digit Recognizer and 85.46% test accuracy for Twitter Sentiment Analysis. The sentiment project reports F1-scores of 0.79 for Negative, 0.87 for Neutral, and 0.87 for Positive.

4.10 Certificates of Completion and Other Communication Mail Proof

The Internship Completion Certificate issued by Codec Technologies Pvt. Ltd. confirms completion of the one-month AICTE & ICAC-approved internship program as an Artificial Intelligence Intern from 01/06/2026 to 02/07/2026. The internship offer letter is included as supporting documentation.

Supporting Document – Internship Offer Letter



INTERNSHIP OFFER LETTER

Dear Hardik Rathour,

We are pleased to offer you a Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: pan India, Hybrid
- Duration : 30/06/2026 to 30/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue -



Dr. Anurag Shrivastava
Talent Acquisition Manager



Google for Education
Partner





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Chandivali IT hub, Mumbai

5. BIBLIOGRAPHY / REFERENCES

- [1] Codec Technologies Pvt. Ltd. Internship Offer Letter, issued 01/06/2026.
- [2] Codec Technologies Pvt. Ltd. Internship Completion Certificate, covering 01/06/2026 to 02/07/2026.
- [3] IILM University. Internship Report Format updated 26-27.